

ABE REVIEW

Q&A IN SOIL AND WATER CONSERVATION ENGINEERING

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Easy Questions

1. Slope of the upstream face of the embankment.

- A. Downstream slope B. Outside slope C. Inside slope D. Upstream slope

Answer: C

2. Inside bottom or sill of the conduit.

- A. Invert B. Inside base C. Inside sill D. Bottom sill

Answer: A

3. Closed conduit designed to convey canal water in full and under pressure running condition, to convey canal water by gravity under roadways, railways, drainage channels and local depressions.

- A. Close siphon B. Pressurized conduit C. Siphon D. Inverted siphon

Answer: D

4. _____ water requirement is the amount of water required in lowland rice production which includes water losses through evaporation, seepage, percolation and land soaking.

- A. Land preparation B. Irrigation C. Crop D. Field

Answer: A

5. _____ water requirement is the amount of water required in lowland rice production which is a function of the initial soil moisture and the physical properties of the soil.

- A. Land preparation B. Irrigation C. Crop D. Land soaking

Answer: D

6. Spacing between irrigation laterals.

- A. Ditch spacing B. Lateral spacing C. Horizontal spacing D. Irrigation spacing

Answer: B

7. Deep percolation of water beyond the root zone of plants, resulting in loss of salts or nutrients.

- A. Vertical percolation B. Root zone percolation C. Leaching D. Salt leaching

Answer: C

8. Canal with impermeable material (usually concrete) for channel stabilization and/or reduced seepage.

- A. Line canal B. Lined canal C. Unlined canal D. Impermeable canal

Answer: B

9. Allowable pollutant-loading limit per unit of time, which the wastewater generator is permitted to discharge into any receiving body of water or land.

- A. Pollutant limit B. Loading limit C. Allowable limit D. Wastewater limit

Answer: B

10. Portion of the pipe network between the mainline and the laterals.

- A. Diversion pipe B. Manifold C. Main-lateral pipe D. Reducer

Answer: B

11. Spillway which is not excavated such as natural draw, saddle or drainage way.

- A. Surface spillway B. Flood spillway C. Natural spillway D. Earth spillway

Answer: C

12. Constant flow depth along a longitudinal section of a channel under a uniform flow condition.
- A. Critical depth B. Constant depth C. Laminar depth D. Normal depth

Answer: D

13. Maximum elevation of the water surface which can be attained by the spillway-type dam or reservoir without flow in the spillway.
- A. Normal storage B. Maximum Storage C. Critical elevation D. Design depth

Answer: A

14. Maximum elevation of the water surface which can be attained in an open channel without reaching the freeboard.

A. Normal storage B. Maximum Storage C. Critical elevation D. Design depth

Answer: D

15. In what condition is the open channel freeboard used for water conveyance?

A. Maximum flow B. Emergency flow C. Inundation D. Rainy days

Answer: C

16. The primary purpose in limiting water flow not to go below minimum velocity.

A. Avoid percolation B. Avoid sedimentation C. Avoid critical depth D. Optimize flow

Answer: B

17. Open channel flow is water flow that is conveyed in such a manner that top surface is exposed to the atmosphere such as flow in canals, ditches, drainage channels, culverts, and pipes under _____ flow conditions.

A. Full B. Partially full C. Normal D. Critical

Answer: B

18. Part of the system that impounds the runoff.

- A. Storage B. Reservoir C. Impounding D. Runoff collector

Answer: B

19. Slope at the downstream face of the embankment.

- A. Outside slope B. Inside slope C. Side slope D. Soil gradient

Answer: A

20. Ratio between reference evapotranspiration and water loss by evaporation from an open water surface of a pan.

- A. Pan coefficient B. Evaporation ratio C. Reference pan ratio D. ET ratio

Answer: A

21. Rate of water loss by evaporation from an open water surface of a pan.

- A. Surface evaporation B. Sunken evaporation C. Pan evaporation D. Evaporation loss

Answer: C

22. Vertical flow of water below the root zone which is affected by soil structure, texture, bulk density, mineralogy, organic matter content, salt type and concentration.

- A. Leaching B. Percolation C. Infiltration D. Seepage

Answer: B

23. Vertical flow of water to carry salts contained in water.

- A. Leaching B. Percolation C. Infiltration D. Seepage

Answer: A

24. Method to determine the rate of flow under laminar flow conditions through a unit cross sectional area of soil under unit hydraulic gradient.

- A. Permeability test B. Laminar test C. Flow test D. Hydraulic test

Answer: A

25. The process by which the soil is removed from its natural place.

- A. Soil removal B. Runoff C. Soil erosion D. Leaching

Answer: C

26. A kind of terrace which consists of a series of flattened areas.

- A. Broad-base terrace B. Bench terrace C. Conservation terrace D. Rice terrace

Answer: B

27. The practice where legumes are plowed or incorporated into the soil.

- A. Legume incorporation B. Legume manuring C. Green manuring D. Manuring

Answer: C

28. Farming practice where plowing and harrowing are done along the contour.

- A. Contouring B. Strip cropping C. Crop row aligning D. Contour plowing

Answer: A

29. The simplest method of determining soil erosion over a period of time.

- A. Catchment method B. Erometer method C. Plumb bob method D. Pin method

Answer: D

30. Advanced form of erosion.

- A. Rill erosion B. Gully erosion C. Sheet erosion D. Advanced erosion

Answer: B

31. Dam which resists water flow of water by its weight.

- A. Resisting dam B. Buttress dam C. Gravity dam D. Arc dam

Answer: C

32. Dam consisting of stones enclosed in cyclone wires which allows water passage.

- A. Gabion dam B. Stone dam C. Cyclone dam D. Interlink dam

Answer: A

33. Material used to cover the soil to minimize evapotranspiration.

- A. Plastic B. Mulch C. Leaves D. Soil cover

Answer: B

34. Geological formation shaped by the dissolution of a layer or layers of soluble bedrock, usually carbonate rocks such as limestone or dolomite.

- A. Geological layer B. Dissolved layer C. Soluble layer D. Karst topography

Answer: D

35. Scientific name of carabao grass commonly used in vegetated open channels.

- A. *Glerisedia sepium* B. *Paspalum conjugatum* C. *Cyperus rotundos* D. *Leucaena leucocephala*

Answer: B

Moderate Questions

36. Philippine geographical constant for determining terrace vertical interval.

- A. 0.5 B. 0.8 C. 1 D. 1.5

Answer: A

37. Constant for soil erodibility and cover conditions for erodible soil and poor cover.

- A. 0.5 B. 0.8 C. 1 D. 1.5

Answer: C

38. Estimate the peak runoff rate in cubic meters per second from a 20-hectare drainage basin if rainfall depth for 6 hours reached 60 mm. Use typical runoff coefficient of 0.35.

- A. 2 B. 0.02 C. 0.2 D. 2.2

Answer: C

Solution:

Determining rainfall intensity I

$$I = 60 \text{ mm}/6 \text{ hrs} = 10 \text{ mm/hr}$$

Determining Peak Runoff Rate, q

$$q = 0.0028 \text{ CIA} = 0.0028 (0.35)(10 \text{ mm/hr})(20 \text{ has}) = 0.2 \text{ cubic meters/second}$$

39. Estimate the runoff volume in cubic meters from a 20-hectare drainage basin if rainfall depth for 6 hours reached 60 mm and duration of runoff is about 3 hours. Use typical runoff coefficient of 0.35.

- A. 1,080 B. 1,800 C. 1,008 D. 8,100

Answer: A

Solution:

Determining rainfall intensity I

$$I = 60 \text{ mm}/6 \text{ hrs} = 10 \text{ mm/hr}$$

Determining Peak Runoff Rate, q

$$q = 0.0028CIA = 0.0028 (0.35)(10 \text{ mm/hr})(20 \text{ has}) = 0.2 \text{ cubic meters/second}$$

Determining Total Runoff Volume, Q

$$\begin{aligned} Q &= 0.5 \times \text{Peak Runoff Rate} \times \text{Duration} \\ &= 0.5qT \\ &= 0.5 (0.2 \text{ cu.m/s}) (3 \text{ hrs} \times 3600 \text{ s/hr}) \\ &= 1,080 \text{ cubic meters} \end{aligned}$$

Difficult Questions

40. A trapezoidal concrete gravity dam has vertical upstream wall. Water depth in the upstream side is 50 meters above the dam base. Freeboard of 15% based on design depth. What is the force of water exerted against dam per meter of dam width at deepest section?

A. 1,100 tonnes B. 1,150 tonnes C. 1,200 tonnes D. 1,250 tonnes

Answer: D

Solution:

Determining the force of water P exerted against dam per meter of dam width at the deepest section, P

$$P = 0.5 \times \text{Water density} \times H^2 = 0.5 \times 1 \text{ tonne/m}^3 \times (50 \text{ m})^2 = 1,250 \text{ tonnes}$$

41. A trapezoidal concrete gravity dam has vertical upstream wall. Water depth in the upstream side is 50 meters above the dam base. The base of dam is 78 meters along water flow. What is the uplift force of seepage water below the dam per meter of dam width at deepest section?

A. 1,900 tonnes B. 1,950 tonnes C. 2,000 tonnes D. 2,050 tonnes

Answer: B

Solution:

Uplift = $0.5 \times \text{Water density} \times \text{Depth} \times \text{Base}$

$$\text{Uplift} = 0.5 \times \text{Water density} \times H \times B = 0.5 \times 1 \text{ tonne/m}^3 \times 50\text{m} \times 78\text{m} = 1,950 \text{ tonnes}$$

42. A trapezoidal concrete gravity dam has inclined upstream wall. Resisting moment relative to the dam toe is 263,953.2 tonne-meters while overturning moment is 96,233.3 tonne-meters per meter of dam width at deepest section. What is the safety factor against overturning? Is it safe?

A. 0.36, unsafe B. 2.74, safe C. 3.96, safe D. 5.48, excessively safe

Answer: B

Solution:

Determining the Factor of Safety against Overturning, FS_o

$$FS_o = RM/OM = 263953.2/96233.3 = 2.74$$

Since $FS_o \geq 2$, the dam is safe from overturning.